

JianKui Zhou

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Education

Fudan University, Shanghai, China | 2024.09 - Now

Ph.D.student in Computer Science (collaboration with MSRA)

Thesis: AI Value Alignment

ShanghaiTech University, Shanghai, China | 2021.09 - 2024.06

M.S. in Computer Science (advisor: Professor YueQiu)

Thesis: Stein variational gradient descent

HuaZhong Agricultural University, Wuhan, China | 2016.09 - 2020.06

B.Eng. in Economics

Thesis: Social complex network computing.

Technology and Major Courses

- **Technology:** python, pytorch, matlab, c++
- **Major Courses:** High dimensional probability, Convex optimization, Computer vision, Algorithm design

Teaching Assistant

Linear algebra in Information Science | 2021.9 - 2022.01

- Prof.Manolis is the teacher of this course. Teaching advanced algebra, algebraic geometry, group structure in matrix completion, spectral clustering, PCA and other aspects of theoretical analysis.

Scientific research and Project experience

Machine Learning | 2022.12 - Now

- The theoretical research direction involves analyzing the variance collapse caused by Stein's method. In practical engineering, the focus is on replicating protein structure prediction based on DDPM in generative models.
- In addition, knowledge of theoretical machine learning and familiarity with basic theories such as PCA, ICA, SVM, and Kalman filtering are required.
- Analysis has been done on theoretical and related experimental aspects of matrix completion and the low-rank manifold structure of data.
- Familiarity with convex optimization algorithms and intelligent optimization algorithms such as genetic algorithms is required.

Bayesian inference | *Stein variational gradient descent* 2022.02 - 2023.01

- An efficient Bayesian inference method using particles was developed, which avoids the curse of higher dimensions in the Stein method. The upper bound of variance collapse was theoretically determined, and the finiteness of the inferred sample covering sample space was proven. paper

AI Alignment | *About Alignment Tax* 2024.09 - 2025.04

- Developed an alignment tax benchmark designed to standardize the evaluation of various issues that arise after model alignment.
- Completed a paper for AAIL-26. The data and evaluation methodology have been open-sourced. Huggingface

Internship Experience | *Microsoft Research Asia (MSRA)* 2023.04 - 2024.04

- Conduct research on state space models and pioneer their application in spatial relationship modeling.
- The model developed in the research addresses the issue of inaccuracies in long-term forecasting by combining the Gumbel-Sinkhorn method for spatial relationship sorting with the Transformer. The model achieves state-of-the-art results on six benchmarks.
- Monkey EEG Data Analysis to Determine Brain Region Relationships.